Report of Assignment

Course name: High-Dimensional Data(Lab)
Course No: STA 05425188

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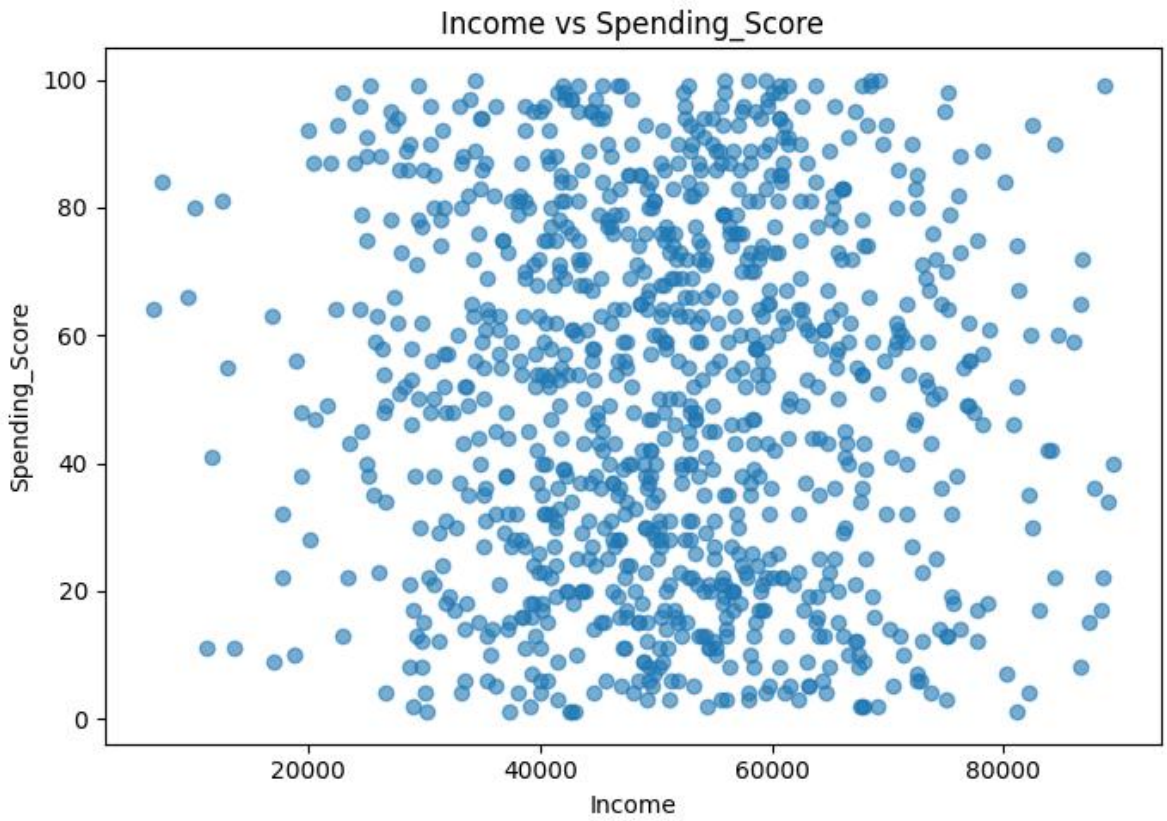
1)

(a) Scatter plots: Income vs Spending_Score

(i) Income and Spending_Score

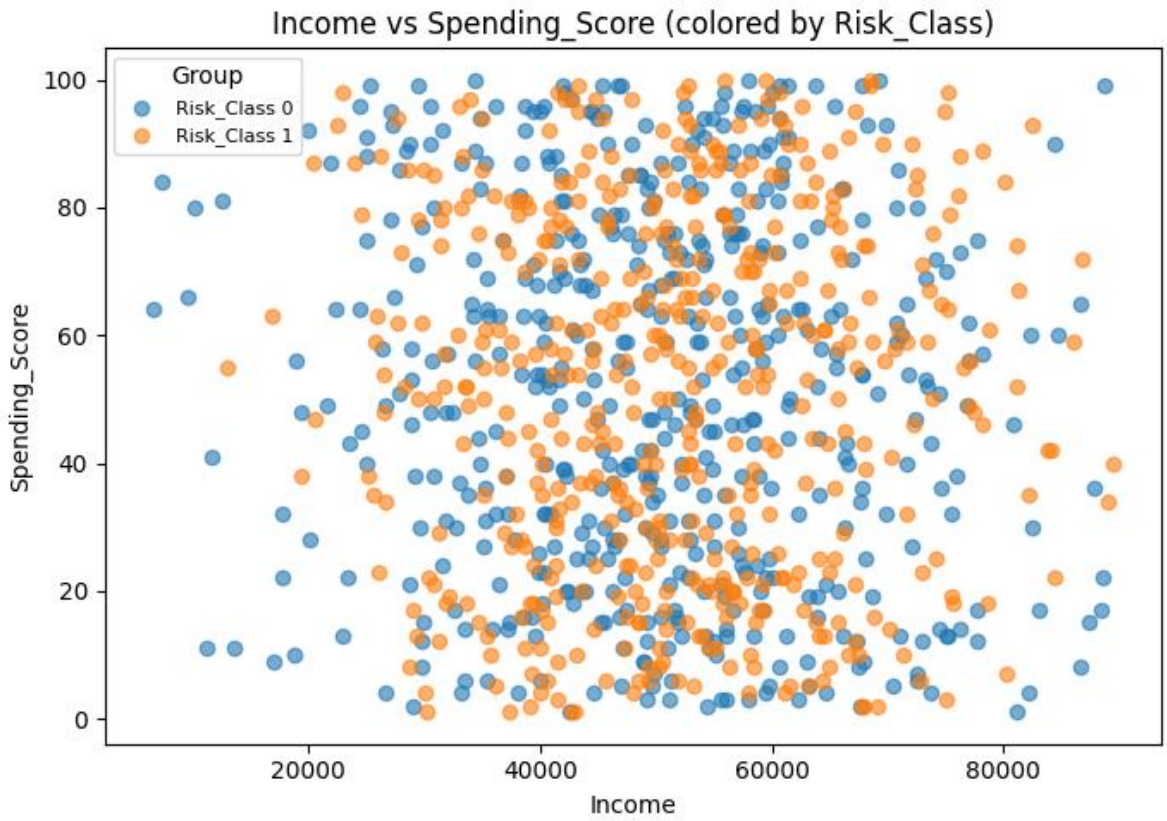


What the scatter shows: the points are widely spread with **no clear linear trend**—customers at similar incomes can have very different spending scores.

Quick numeric check: the correlation is about **-0.04**, which is essentially *no relationship*.

Interpretation: In this dataset, **Income alone is not a strong predictor of Spending_Score**. Spending behavior likely depends more on other factors (preferences, segment, promotions, etc.).

(ii) Income and Spending_Score colored by Risk_Class



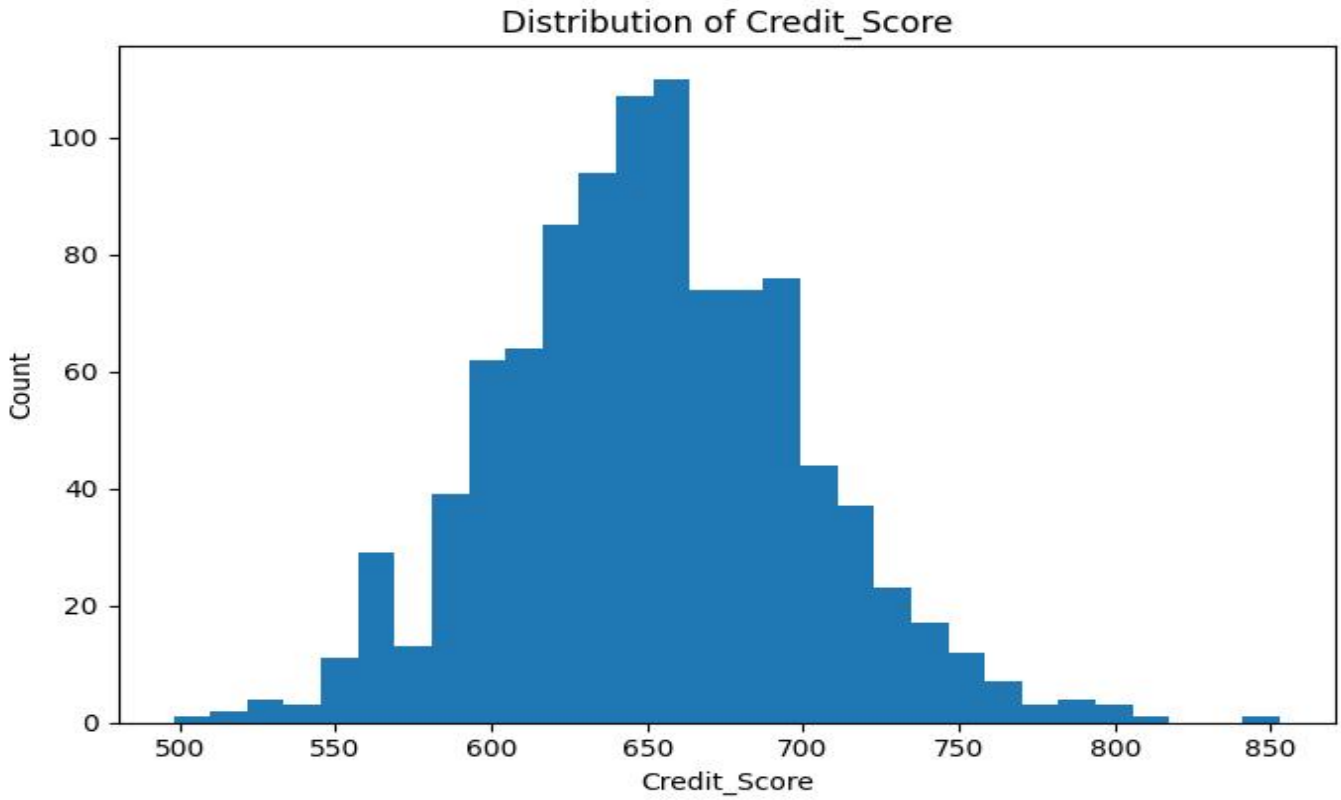
What the colored scatter shows: the two Risk_Class groups largely **overlap** across both income and spending score.

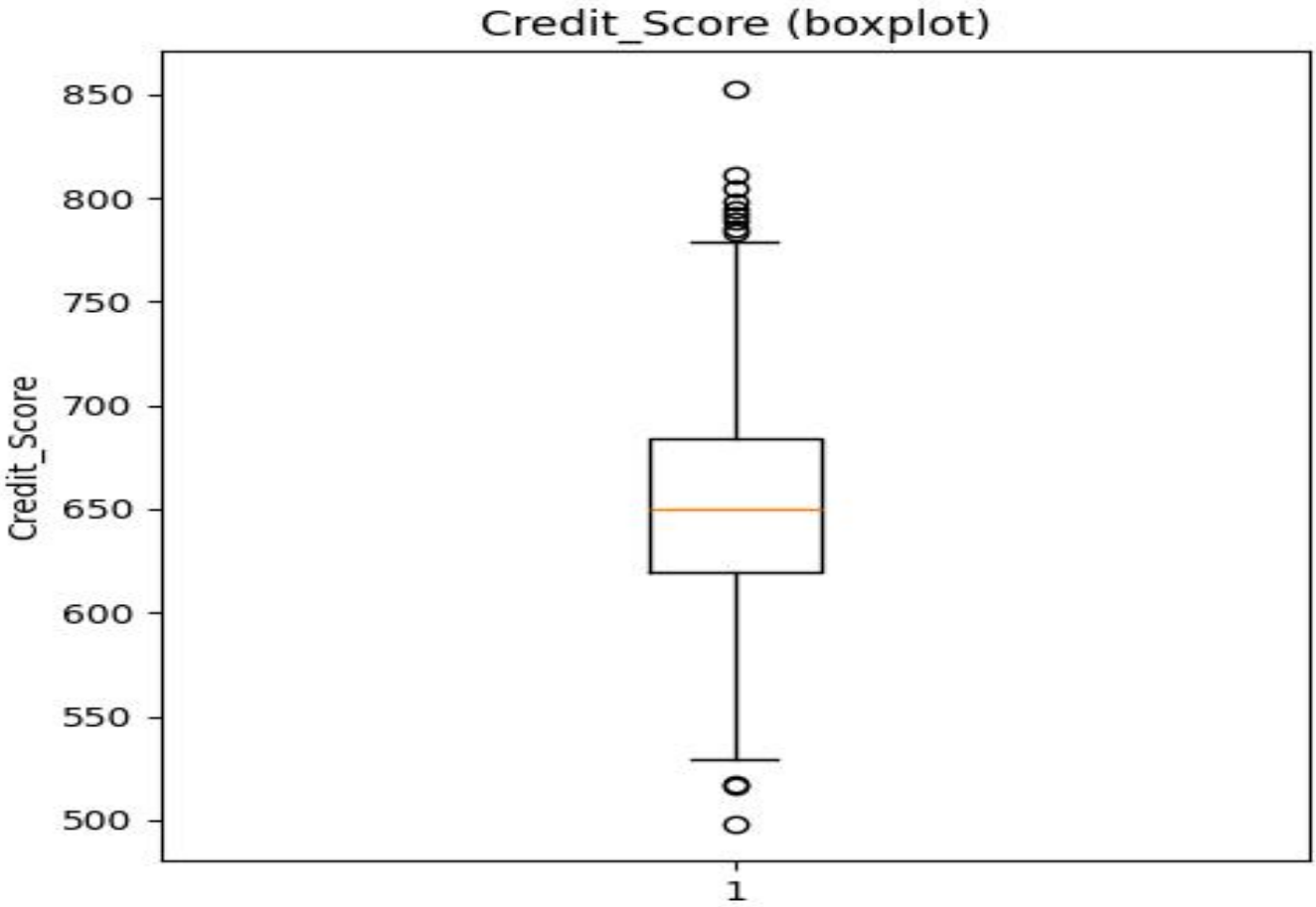
Group averages (Income, Spending_Score): they are **very similar** across Risk_Class values.

Interpretation: Risk_Class does not visibly separate customers in the Income–Spending space here. If Risk_Class is meaningful, it may be driven by other features (e.g., fraud risk, credit behavior), not income/spending directly.

(b) Distributions

(i) Credit_Score distribution


Distribution of Credit_Score
Count
Credit_Score
0
20
40
60
80
100
500
550
600
650
700
750
800
850


Credit_Score (boxplot)
Credit_Score
500
550
600
650
700
750
800
850
1

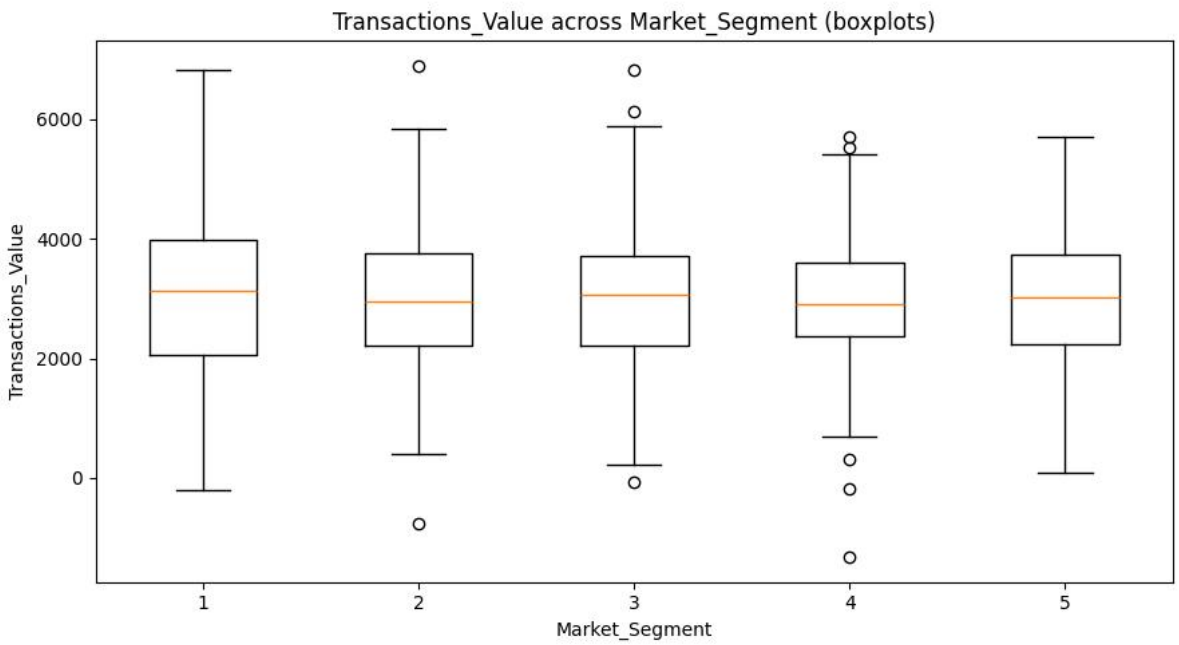
What the histogram + boxplot show:

- Centered around roughly **650** (median ≈ 650).
- Spread is moderate (most values roughly in the **620–684** range).
- Slight right tail (a few higher scores), but **not strongly skewed**.

Interpretation: Credit scores are **fairly tightly clustered** around the mid-600s, suggesting most customers have "average-to-good" credit in this sample, with fewer extremes.

(ii) Transactions_Value across Market_Segment

What the Market_Segment boxplots show:

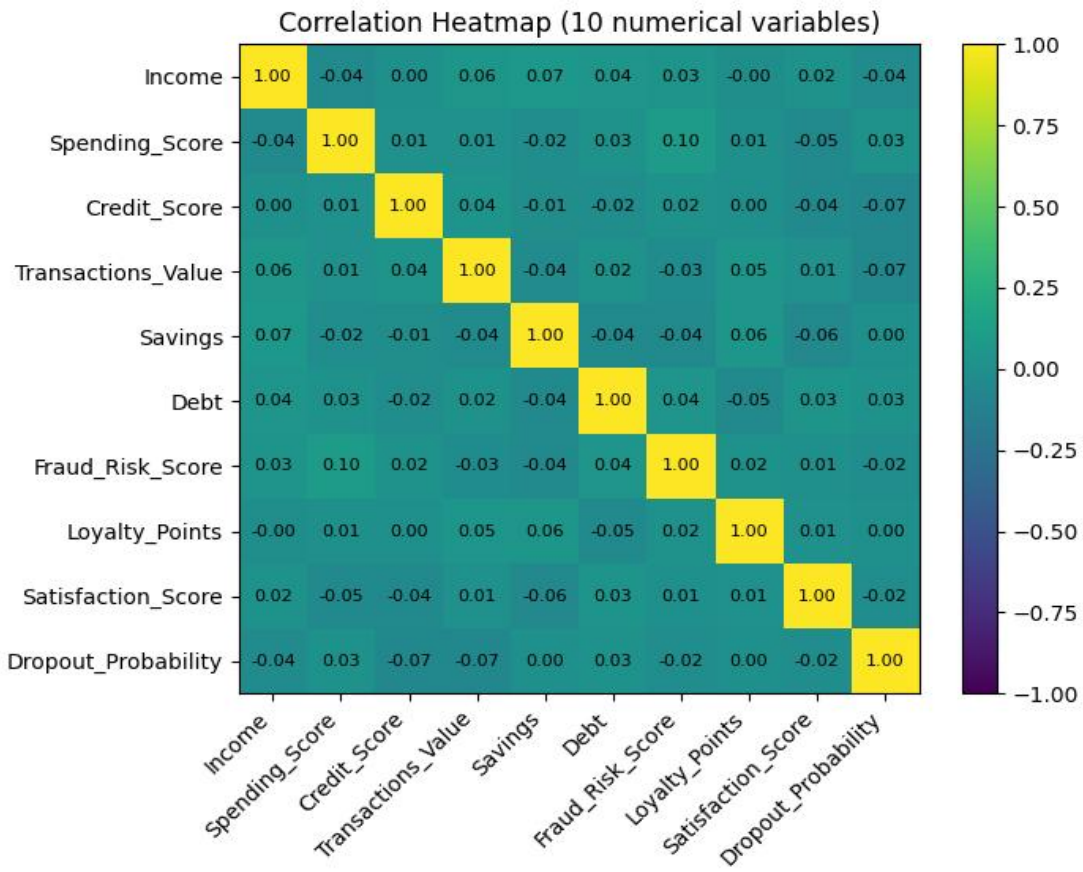


- The **medians are very similar** across segments (around ~2900–3150).
- Segments differ a bit in **spread and outliers**.
- Some segments show **more extreme low/high values** (including a few unusually low/negative values, which may indicate refunds/returns or data quirks).

Interpretation: Market_Segment does **not strongly shift typical transaction value**, but it may relate to **volatility/outliers** (some segments have wider variability).

(c) Correlation heatmap (10 numerical variables) + strong relationships

Using these 10 variables:
Income, Spending_Score, Credit_Score, Transactions_Value, Savings, Debt, Fraud_Risk_Score, Loyalty_Points, Satisfaction_Score, Dropout_Probability



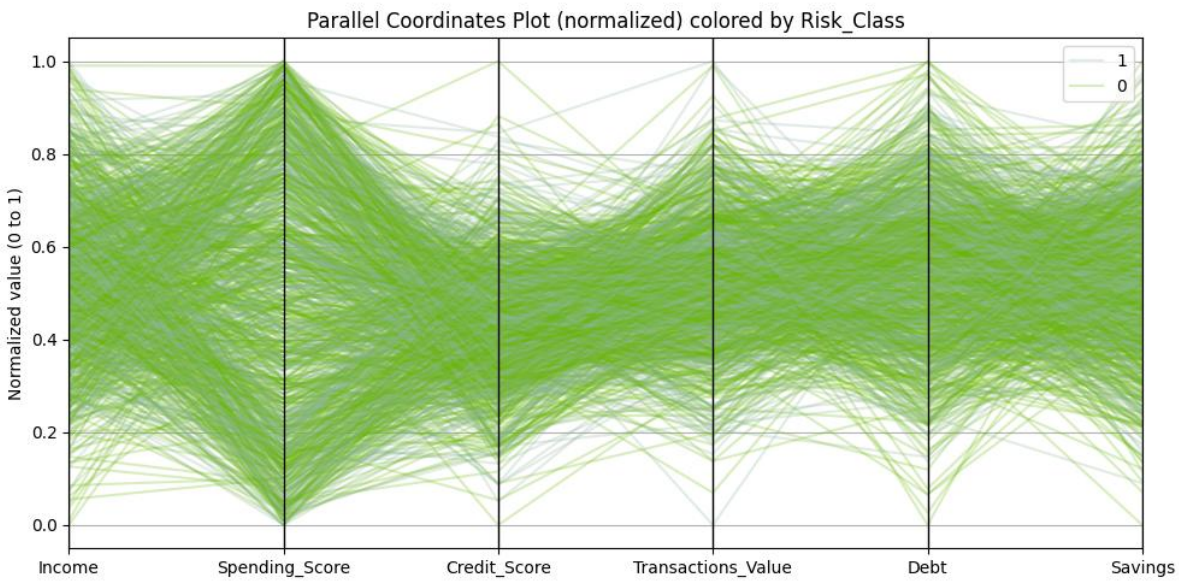
Important note

The correlations in this dataset are **generally weak** (none close to "strong" like $|r| \geq 0.5$). So the "strongest" relationships are **strong only relative to the rest**.

Top 3 strongest relationships (by absolute correlation)

1. **Spending_Score ↔ Fraud_Risk_Score**: **r ≈ +0.10**
   Interpretation: Higher spending score is *slightly* associated with higher fraud risk score, but the effect is very small.
2. **Credit_Score ↔ Dropout_Probability**: **r ≈ -0.07**
   Interpretation: Better credit is *slightly* associated with lower dropout probability (again, weak).
3. **Transactions_Value ↔ Dropout_Probability**: **r ≈ -0.07**
   Interpretation: Higher transaction value is *slightly* associated with lower dropout probability, but not strongly.

2)



(a) Curse of dimensional visualization (overplotting + interpretability)

- **Overplotting:** Each customer becomes a polyline. With many customers, lines overlap heavily, creating a "hairball," hiding patterns and making class differences hard to see.
- **Interpretability drops as dimensions grow:** With more numerical variables (axes), the plot becomes visually busy. It's hard to track individual lines across many axes, and the viewer struggles to identify consistent trends.
- **Axis-order dependence:** Patterns can look different if you reorder variables, so conclusions can become subjective.

(b) Why it's useful, but gets hard as observations increase

- **Useful:** It's one of the few plots that can show multiple variables at once, letting you see:
  - potential separation between groups (Risk_Class),
  - unusual profiles (outliers),
  - rough multivariate "shapes" (e.g., high income + low debt + high credit).
- **Hard with many observations:** As sample size grows, the number of lines grows, and:
  - overlaps increase,
  - dense regions become opaque,
  - outliers may disappear in clutter,
  - class coloring becomes less meaningful because everything blends.

(c) Motivation for dimensionality reduction

Because high-dimensional plots become cluttered and hard to interpret, we often compress the data into **fewer dimensions** while preserving structure:

- Techniques like **PCA** (linear) or **t-SNE/UMAP** (nonlinear) create **2D/3D representations** that are easier to visualize.
- This helps reveal **clusters, separations, and outliers** more clearly than trying to "look" at 5–20 axes simultaneously.

3)

(a) Standardize numeric variables + perform PCA, and why standardization is necessary

Standardization + PCA (what was done)

1. Selected **numeric predictor variables** and excluded:
   - the target: `Dropout_Probability`
   - the label used for coloring: `Risk_Class`
   - coded categorical columns: `Market_Segment`, `Region_Code`, `Channel _ Preference`
2. Standardized the predictors using z-score scaling:

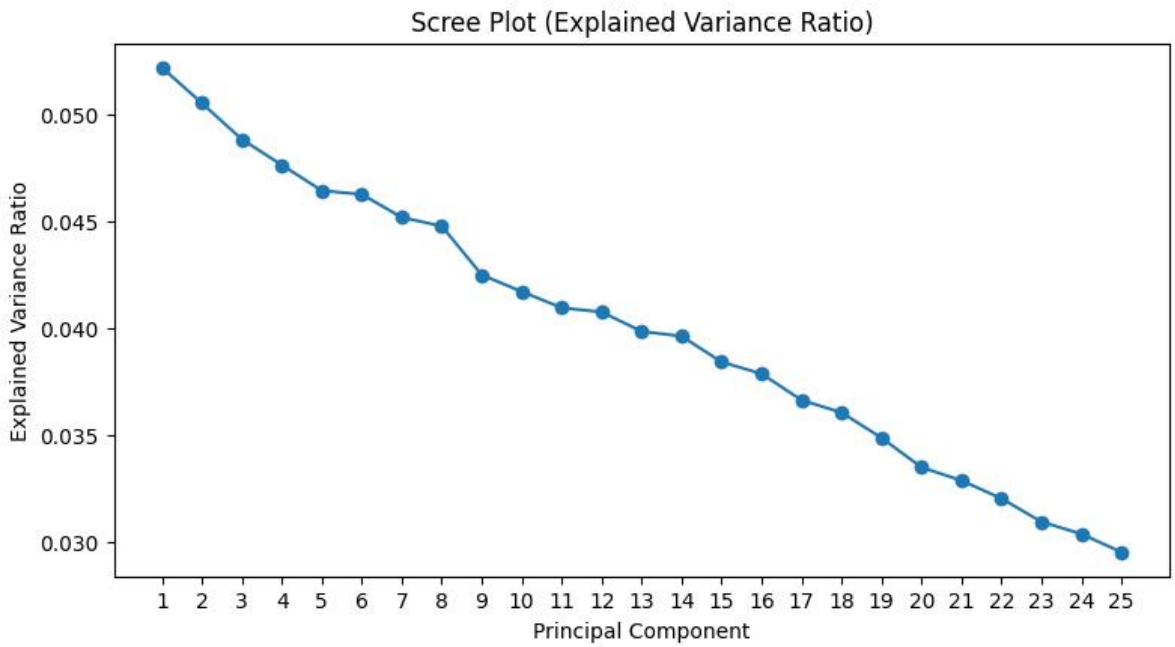
$$z = \frac{x - \mu}{\sigma}$$

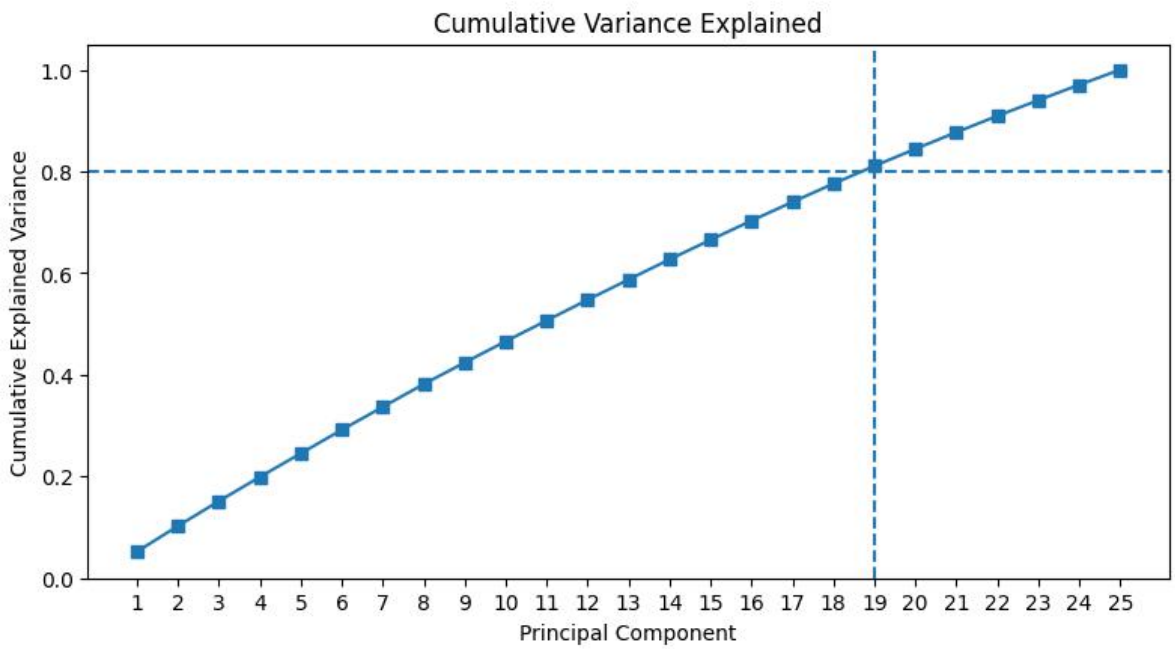
3. Applied PCA on the standardized matrix.

Why standardization is necessary before PCA

PCA is based on the **variance–covariance structure** of the features. If variables are not standardized, those with larger scales (e.g., `Income` in tens of thousands) will have much larger variances and will **dominate** the principal components. Standardization ensures each variable contributes fairly (each gets variance ≈ 1), so PCA captures the **true correlation patterns** rather than measurement units.

(b) Scree plot and number of PCs for ≥ 80% variance





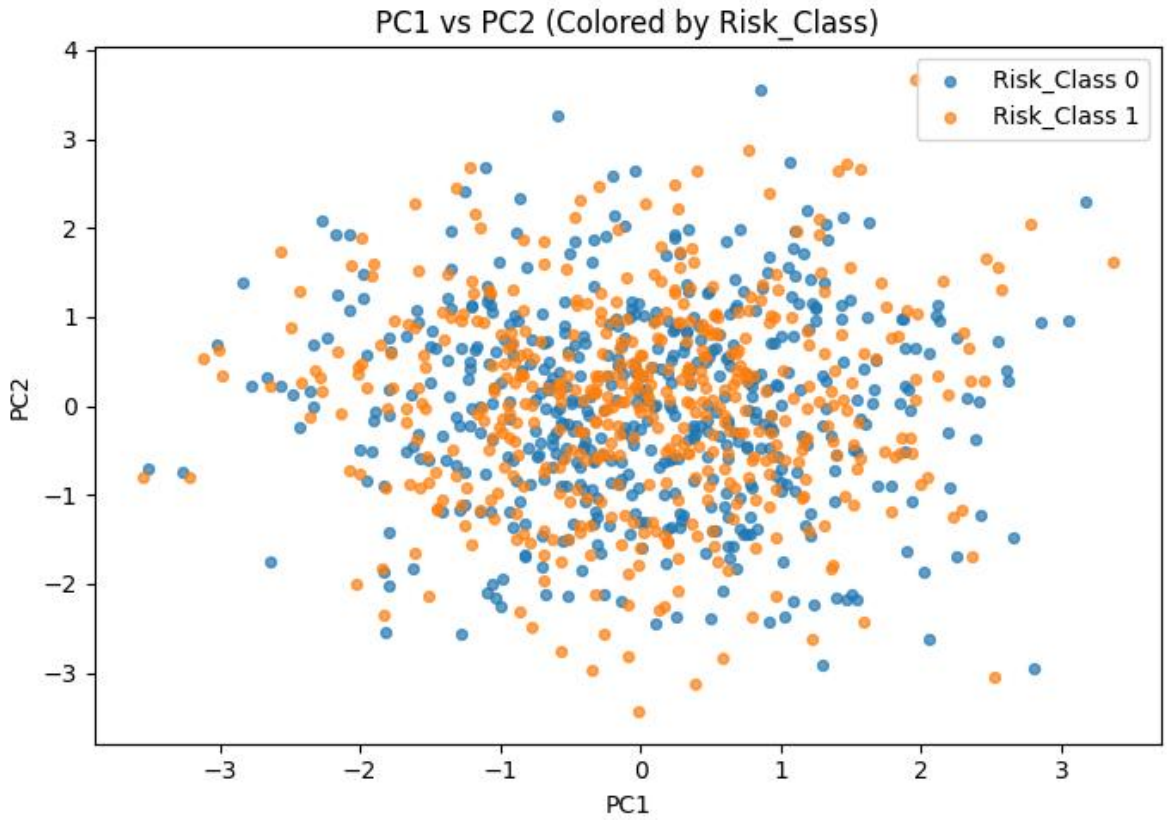
From the PCA explained variance:

- Minimum number of principal components needed to reach **at least 80%** total variance:
  19 principal components
- Cumulative variance explained at PC19:
  ≈ 81.08%

(So, you would retain **19 PCs** if using the "≥80% variance explained" rule.)

(c) PC1 vs PC2 plot colored by `Risk_Class` + interpretation

When plotting the data on the first two principal components (PC1 vs PC2) and coloring by `Risk_Class`:



- The two risk classes show **strong overlap** in the PC1–PC2 space.
- There is **no clear cluster separation** between risk groups using only the first two PCs.

Interpretation:
The largest variance directions (PC1 and PC2) represent major patterns in customer behavior/features, but those patterns do **not** align strongly with `Risk_Class`. Any separation (if it exists) may require:

- higher PCs,
- non-linear methods (e.g., t-SNE/UMAP),
- or supervised dimensionality reduction.

(d) Variables contributing most to PC1 (loadings) + why they matter

Using the **absolute value** of PC1 loadings (largest = most influence on PC1), the top contributors to **PC1** were:

1. **Ad_Clicks** (loading ≈ **-0.367**)
2. **Email_Open_Rate** (≈ **-0.367**)
3. **Spending_Score** (≈ **+0.362**)
4. **Fraud_Risk_Score** (≈ **+0.312**)
5. **Purchases_Month** (≈ **+0.284**)
6. **Discount_Usage** (≈ **-0.254**)
7. **Debt** (≈ **+0.251**)
8. **Website_Visits** (≈ **+0.228**)
9. **Satisfaction_Score** (≈ **-0.206**)
10. **Social_Media_Usage** (≈ **-0.180**)

What this means (importance)

PC1 is the **dominant source of variation** in the dataset. Since its strongest loadings are on engagement and transaction behavior variables (ads/email engagement, spending, purchases, web visits, fraud risk, debt), PC1 mainly represents a **behavior/engagement intensity** dimension rather than demographics (like age/education).

(e) PCR using selected PCs to predict `Dropout_Probability` + why PCR is preferred sometimes

PCR model (what was done)

- Used the selected number of components from (b): **19 PCs**
- Pipeline:
  StandardScaler → PCA(19) → Linear Regression
- Goal: predict `Dropout_Probability`

Example performance (train/test split)

In this dataset, PCR performance was similar to ordinary linear regression, and both were weak (R^2 slightly negative), meaning the linear relationship from these predictors to `Dropout_Probability` is not strong in the current feature set / linear form.

Why PCR may be preferred over ordinary linear regression in high-dimensional settings

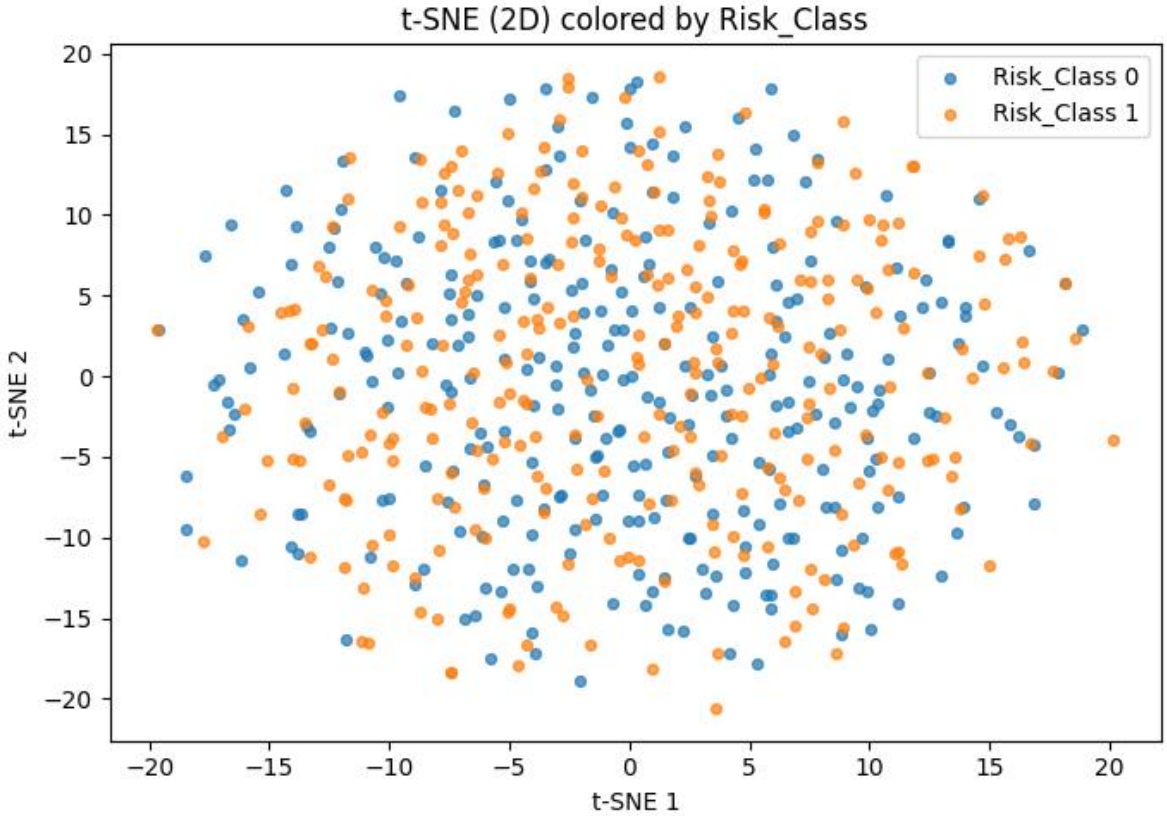
PCR is often preferred when:

- predictors are **highly correlated (multicollinearity)** → OLS coefficients become unstable.
- there are **many predictors** → higher risk of overfitting and noisy coefficient estimates.
- you want to regress on **orthogonal (uncorrelated) components**, improving numerical stability.

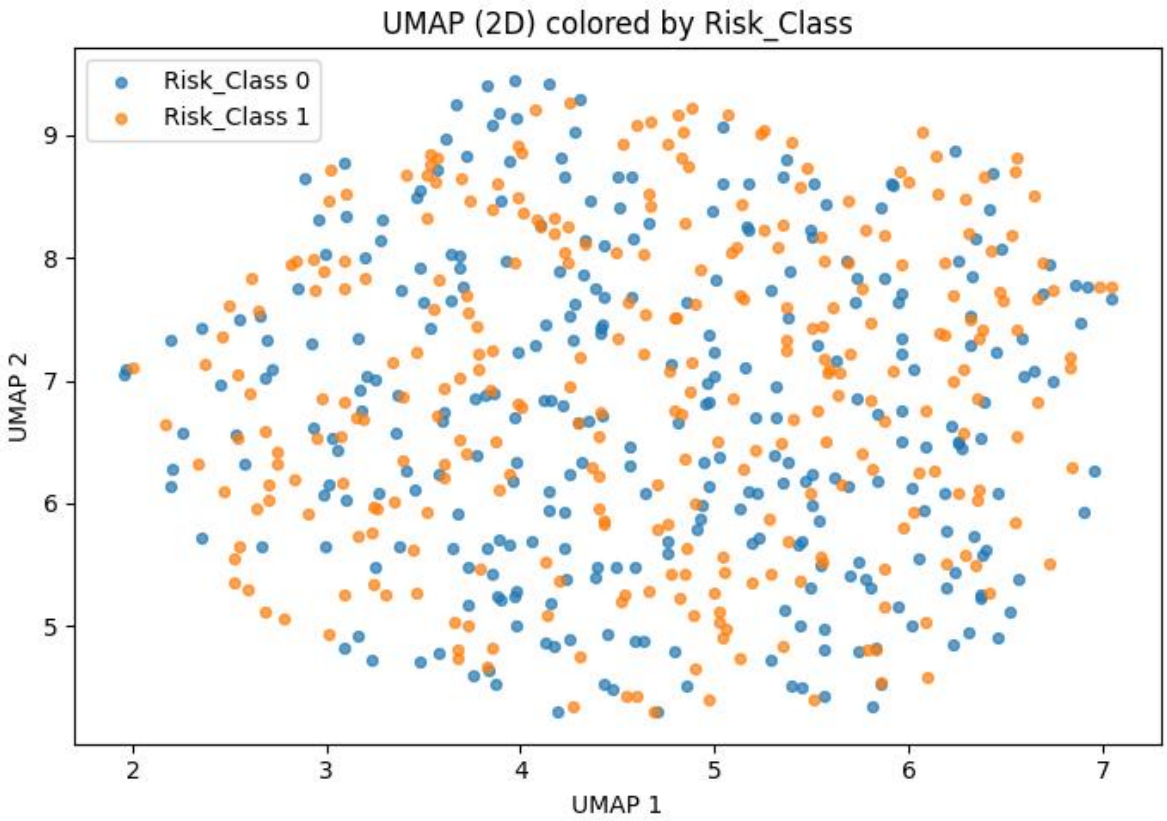
Key idea: PCR reduces dimensionality and mitigates multicollinearity by fitting regression on a smaller set of uncorrelated principal components instead of the original variables.

4)

(a)



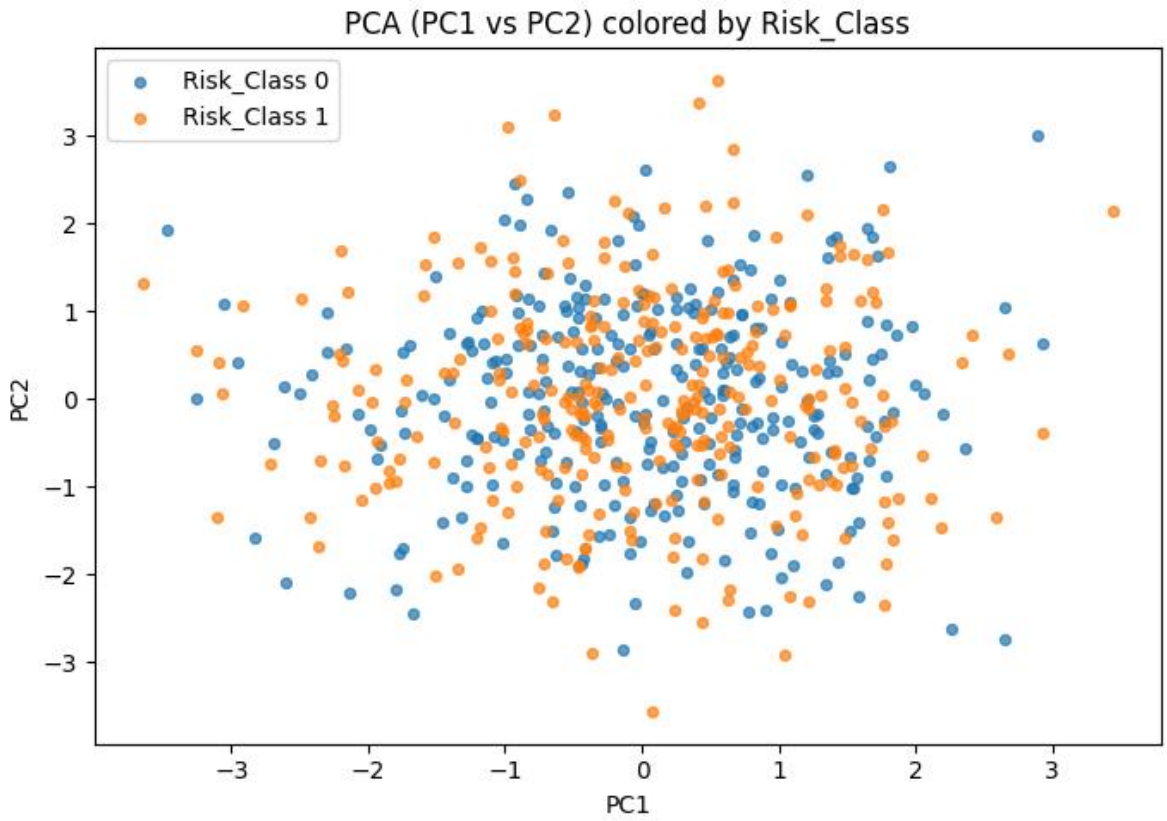
(b)



UMAP vs t-SNE (how to compare what you see)

- If **both** show overlapping colors → little/no separation between risk classes in 2D.
- If **UMAP shows clearer grouping** than t-SNE: UMAP often keeps a bit more *global* structure while still separating local neighborhoods.
- If **t-SNE shows tighter "islands"**: that's common; t-SNE tends to emphasize local clustering (sometimes even when global structure is not meaningful).

(c)



PCA vs t-SNE vs UMAP

i) Interpretability

- **PCA:** High (linear; you can interpret PCs using **loadings**).
- **t-SNE:** Low (nonlinear; axes have no direct meaning; no simple feature loadings).
- **UMAP:** Low–Medium (still nonlinear; axes not directly interpretable; some structure interpretation possible but not via simple loadings like PCA).

ii) Structure preservation (local vs global)

- **PCA:** Best at preserving **global linear** structure (large-scale variance directions).
- **t-SNE:** Best at preserving **local neighborhoods**; global distances/cluster spacing are **not** reliable.
- **UMAP:** Strong **local** preservation and usually **better global** organization than t-SNE (often keeps "overall shape" more).

iii) Computational complexity (practical)

- **PCA:** Fast (scales well; generally the cheapest).
- **t-SNE:** Usually the slowest (can be expensive; often needs subsampling for large data).
- **UMAP:** Typically faster than t-SNE and scales better; often feasible on larger datasets.

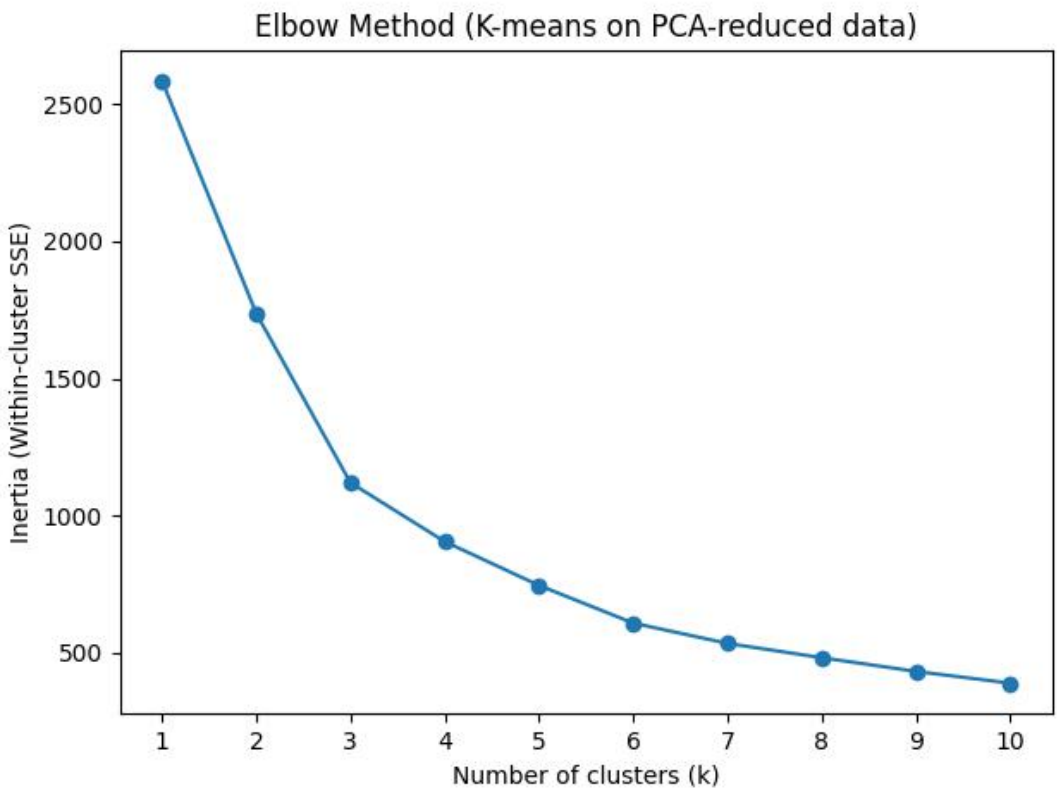
5(a) K-means on PCA-reduced data + Elbow method

What you do

1. **Standardize** features (important for K-means).
2. Reduce to **PCA space** (usually 2 PCs if the task is visualization-driven).
3. Run **K-means** for different values of **k**.
4. Use the **elbow method**: pick the k where inertia stops dropping sharply.

Short quote (what elbow means)

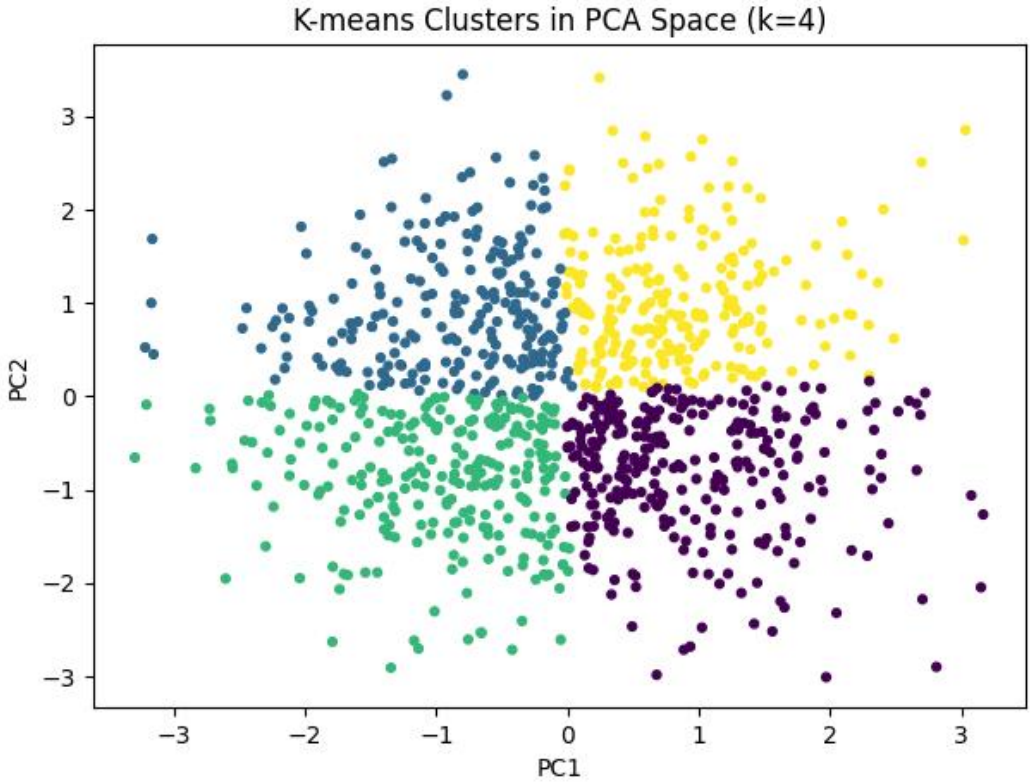
"The idea is to choose k at the point where the inertia starts decreasing more slowly (the 'elbow')."



What to write as justification (ready text)

- The inertia decreases sharply for small k and then the improvement becomes smaller.
- The **optimal k** is chosen at the **elbow point**, where adding more clusters gives **diminishing returns** (small additional reduction in inertia).
- Therefore, we select **k** = ___ (fill from your elbow graph).

(b) Visualize clusters in PCA space + interpret structure



Interpretation (paste under the plot)

- Each point is a customer projected into the **2D PCA space (PC1 vs PC2)**.
- The colors represent **K-means cluster assignments**.
- If clusters appear **well-separated**, it indicates strong segmentation in the customer dataset.
- If clusters **overlap**, it means customers share similar patterns and the boundary is not sharply defined in PCA space.
- Larger clusters indicate **more common customer profiles**, while smaller clusters may represent **niche groups or outliers**.

Optional stronger interpretation (if your teacher wants meaning)

- **PC1** typically represents the direction of maximum overall variation in customer behavior.
- **PC2** captures the second-most important variation pattern.
- Cluster separation along PC1/PC2 implies customer groups differ mainly along those major behavioral/financial dimensions.

6(a) Histogram of `Dropout_Probability` + describe distribution

Interpretation (use this text)

- In this dataset, `Dropout_Probability` ranges from **0.0026 to 0.9999**.
- Mean ≈ **0.494** and median ≈ **0.490**, so it is centered around **0.5**.
- The histogram is **almost uniform** across 0–1 (roughly similar counts in each 0.1 bin), meaning dropout probabilities are **spread evenly** rather than concentrated near 0 or 1.

6(b) Bar chart: average `Dropout_Probability` by `Channel_Preference` (color by channel)

Interpretation (use this text)

Average dropout probability by channel in this dataset:

- Channel 1: **~0.491**
- Channel 2: **~0.483**
- Channel 3: **~0.509**

So, Channel **3** shows the **highest average dropout probability**, while Channel **2** is slightly lower.

6(c) Scatter plot: `Support_Call_Count` vs `Dropout_Probability`, color by `Risk_Class`

Interpretation (use this text)

- This plot shows whether customers with more support calls tend to have higher dropout probability.
- If points trend upward (higher Y as X increases), that suggests **more support calls → higher dropout risk**.
- Coloring by `Risk_Class` helps check whether high-risk customers cluster at higher dropout probability levels.

6(d) Apply clustering/grouping on the scatter plot to identify customer risk-behavior groups

Power BI (two common options)

Option 1: Automatic clustering (if available in your version)

1. Build the scatter chart.
2. Click the visual's **… (More options)** → look for **Find clusters**.
3. Pick number of clusters or auto.

Option 2: Grouping via bins (works in all versions)

1. Create bins for:
   - `Dropout_Probability` (e.g., Low <0.33, Medium 0.33–0.66, High >0.66)
   - `Support_Call_Count` (e.g., Low/High bins)
2. Create a combined group label (e.g., "High Calls – High Dropout").
3. Use that group field as **Legend** in scatter.